

Shared Intelligence for Robot Teleoperation via BMI

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Abstract—This article proposes a novel *shared intelligence* system for brain-machine interface (BMI) teleoperated mobile robots where user's intention and robot's intelligence are concurrent elements equally participating in the decision process. We designed the system to rely on *policies* guiding the robot's behavior according to the current situation. We hypothesized that the fusion of these *policies* would lead to the identification of the next, most probable, location of the robot in accordance with the user's expectations. We asked 13 healthy subjects to evaluate the system during teleoperated navigation tasks in a crowded office environment with a keyboard (reliable interface) and with 2-class motor imagery (MI) BMI (uncertain control channel). Experimental results show that our *shared intelligence* system 1) allows users to efficiently teleoperate the robot in both control modalities; 2) it ensures a level of BMI navigation performances comparable to the keyboard control; 3) it actively assists BMI users in accomplishing the tasks. These results highlight the importance of investigating advanced human-machine interaction (HMI) strategies and introducing robotic intelligence to improve the performances of BMI actuated devices.

Index Terms—Behavior based architecture, brain-machine interface (BMI), motor imagery, neurorobotics, shared intelligence, teleoperation.

I. INTRODUCTION

BRAIN-MACHINE interfaces (BMIs) are communication channels that circumvent the brain's normal output pathways of peripheral nerves and muscles [1]–[3]. A BMI system is designed to acquire, process, and decode neural patterns

and translate them into a control signal for the robotic device. Over the last years, several studies have demonstrated that both healthy and people with disabilities can successfully operate different robotic platforms, including: powered wheelchairs, robotic arms, exoskeletons, telepresence robots [3]–[10]. However, despite the great advances, the usability of brain-actuated robots is still at an early stage [10], especially due to the intrinsic low bit-rate and uncertainty that characterize any BMI system. Indeed, it is still an open challenge how to efficiently transform sparse, discrete, and often unreliable BMI commands to coherent and trustworthy robot's behaviors, in particular, while operating in a complex and uncontrolled environment.

In this scenario, robotic intelligence may play a fundamental role to support the user's intention, to enhance the HMI, and thus, to improve BMI-driven operations. In literature, several studies have shown the potentialities of introducing a sort of robot's intelligence at different levels of interaction with the BMI user. By focusing on works related to teleoperation of mobile robots, two main approaches have been proposed: autonomous robotic behaviors triggered by high-level BMI commands [11]–[14] or the modification of the BMI commands delivered by the user with the surrounding environment perceived by the robot [4], [15]–[18].

In the former, a set of possible final destinations is defined according to the experimental environment and presented by a visual interface to the user. The BMI user commonly selects the desired location (e.g., living room, kitchen) by paying attention to the corresponding stimulus on the screen (e.g., flickering light). Once the destination is selected, the robot autonomously moves to reach it. Although this approach guarantees a low workload for the user (after the selection of the destination, the user is passively carried through the environment), the HMI is sub-optimal, often time-consuming, and definitely not ecological. Furthermore, the approach strongly depends on the *a priori* knowledge of the environment, and thus, any unexpected change could make the system fail. Finally, such interaction modality heavily limits the independence of the user by not allowing full control of the robotic device [19].

The latter approach is known in literature as *shared control* [20], [21] or *shared autonomy* [22], [23] and it provides a more advanced interaction modality between human and robot. The robot is in charge of handling the low-level details for safe navigation (e.g., obstacle avoidance) while the BMI user is allowed to deliver high-level commands at any time to adjust the direction of the motion accordingly to the desired path. It has been already demonstrated that such a semi-autonomous navigation strategy promotes the emergence of expected robot

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behaviors and, as a consequence, the coupling and the cooperation between user and machine [4], [17], [18], [24]. However, even this approach suffers from limitations: typically, the robot's role is extremely limited and hierarchically subordinated to the user's command. Furthermore, the decision process is purely reactive according to predefined procedures that are activated when specific conditions are detected. For instance, most studies in literature exploits robotic intelligence to adjust the current trajectory if potential collisions are detected [15]–[17], [25]–[29] or to assist user in demanding situations (e.g., passage through a door) [15], [30]–[32]. However, the proposed algorithms often rely on strict environment-dependent behaviors and on hard-code constraints that might complicate their replication and validation in different situations.

In this work, we propose a third level of interaction strategy between BMI user and robotic device, namely *shared intelligence*. The key idea of this novel approach is that user's intent—through the BMI system—and robot's intelligence are concurrent elements equally participating in the decision process. Our approach is designed to take advantage of the reactivity of traditional *shared control* algorithms and, at the same time, to exploit the safety and the reduced workload of goal-directed autonomous navigation solutions. The method and the implementation adopted in our *shared intelligence* system are reported in Section II. Section III is devoted to describing the experiment designed to evaluate the new approach with healthy subjects during BMI teleoperation of a mobile robot. In Section IV, we present the experimental results, and in Section V we discuss them in comparison to prior literature, and we propose possible extensions of the work in different BMI robotic applications. Finally, Section VI concludes this article.

II. SHARED INTELLIGENCE SYSTEM

The aim of our *shared intelligence* system is to provide a navigation framework for BMI teleoperated mobile devices where user's and robot's intelligence concurrently contribute to the robot trajectory without any *a priori* information on the environment nor on the user's final destination. To this end, we have exploited the concept of *subgoal* that represents an intermediate destination of the robot in the surrounding area. The identification of the *subgoal* location is determined by a set of *policies* that have been designed by taking inspiration from the artificial potential fields. Each *policy* independently treats a different aspect of the *subgoal* position in the proximity space—such as the distance of the *subgoal* from the robot, the most probable direction, or the proximity to possible targets. Formally, each *policy* is represented by a decision function taking information from heterogeneous sources (i.e., robot's sensors or BMI commands) and outputting a probability grid centered in the robot's reference frame and extending in its near neighborhood. We compute the next *subgoal* position by fusing the output of these independent *policies* and by selecting the location with the highest probability. Finally, the *subgoal* is fed to a traditional motion planner to let the robot autonomously

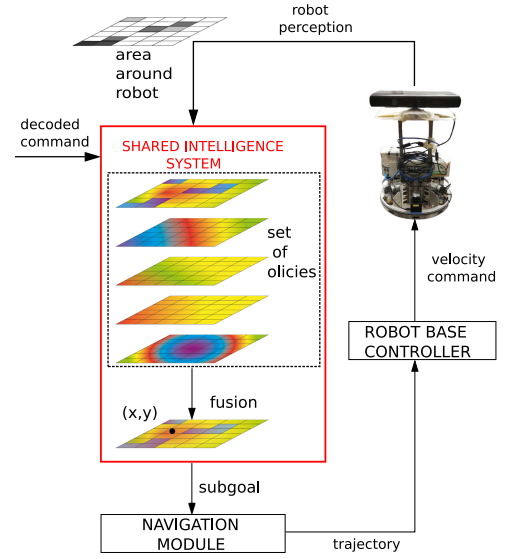


Fig. 1. Schematic representation of our *shared intelligence* system. The user's commands and the information from robot's sensors are equally combined according to a set of *policies*. Each *policy* determines a probability grid centered in the robot's current position. The fusion of all the *policies* determines the next *subgoal* for the robot. Then, the navigation module computes the best trajectory the robot should follow to approach the *subgoal* during the navigation, while the robot base controller is in charge of determining the velocity commands according to the computed trajectory.

decide the best and safest path to reach it. Fig. 1 illustrates a schematic representation of our system.

In the following sections, we will present the *policies* currently implemented in our system and details on the fusion and recomputation of the *subgoal*.

A. Obstacle-Avoidance Policy

The *obstacle-avoidance policy* is introduced to prevent collisions of the robot. In the proposed implementation, the probability that the *subgoal* is located in a given position depends on the distance of the perceived obstacles. Formally, let $\mathcal{A}$ be the area perceived by the robot, d_o the distance of the closest obstacle $\forall (x, y) \in \mathcal{A}$ and s_r the size of the robot, the *obstacle-avoidance policy* p_o is defined as follows:

$$p_o = (1 - e^{-\lambda d_o}) \text{ with } d_o > s_r \quad (1)$$

with $d_o > s_r$ and the smoothing parameter $\lambda > 0$ (see Appendix). As illustrated in Fig. 2, the closer is a position to an obstacle, the higher is its effect and, thus, the smaller is the probability to place the *subgoal* there. It is worth noticing the probability assigned by the *obstacle-avoidance policy* is forced to zero if the positions is occupied by an obstacle.

B. Distance Policy

The *distance policy* defines a range of favourite distances $[d_{min}, d_{max}]$ where the probability to place the *subgoal* is higher. The motivation for the *policy* is to avoid deadlock situations when, for any reason, the robot is unable to move from the current position. Given $\mathcal{A}$ the area around the robot and d_r the distance of the robot $\forall (x, y) \in \mathcal{A}$, we defined the *distance*

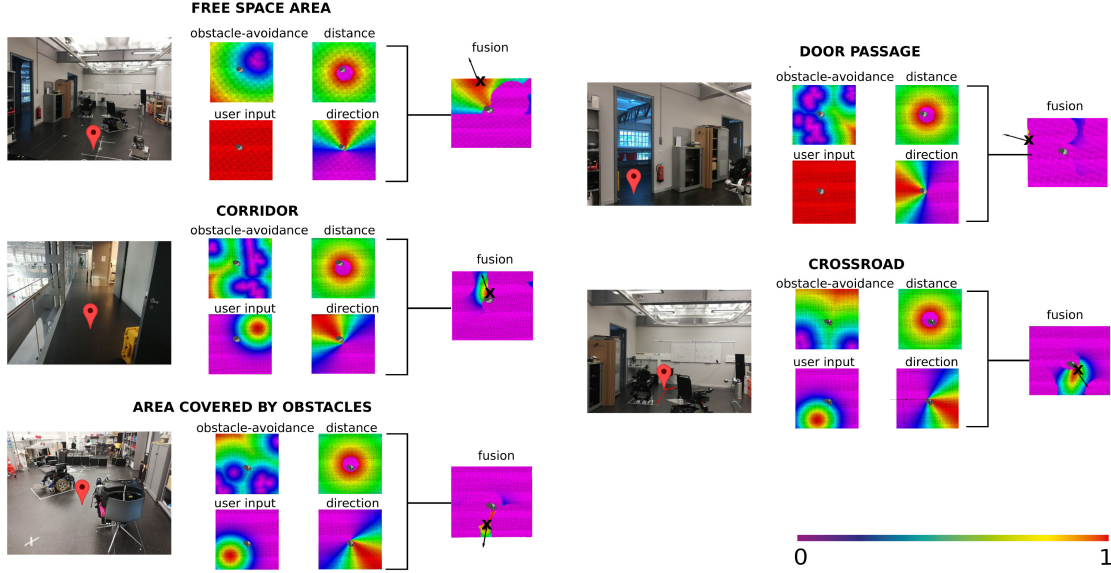


Fig. 2. Illustrative example of application of our *shared intelligence* system in typical situations in robotics: free space area, door passage, corridor, crossroad, and area covered by obstacles. On the left, pictures represent the different situations. In the middle, the top view of the probability grid is computed for each *policy*. The area in red is the most preferable for setting the *subgoal* (high probability), while the ones in fuchsia/blue are the least probable. On the right, the fusion achieved by combining the *policies* is presented from the top perspective. The computed *subgoal* is represented with a black arrow (direction) and a cross (position).

policy p_d as follows:

$$p_d = \kappa \cdot e^{-(d_r - d_{min})} \text{ with } d_{min} \leq d_r \leq d_{max} \quad (2)$$

with d_{min} and d_{max} defining the favourite range (see Appendix) and $\kappa = 1/(Q3 + 1.5IQR) = 1/(\ln 4 + 1.5 \ln 3)/(d_{max} - d_{min})$ to match the Tukey's fences [33] for values greater than d_{max} . Examples of the resulting probability grid are depicted in Fig. 2.

C. User Input Policy

This *policy* increases the probability of a certain region of space to contain the *subgoal* in accordance with the user's input. In order to favor the robot's rotation according to the command delivered by the user, in our current implementation, we create an attractive Gaussian probability distribution respectively on the left or the right (with respect to the robot's heading direction). We defined the *user input policy* p_u as follows:

$$p_u = e^{-\frac{d_u^2}{2\sigma_u^2}} \quad (3)$$

where d_u and $\sigma_u > 0$ are the factors controlling the attractive influence of the user's input. Specifically, d_u is computed as the distance between each cell (x, y) of the probability grid and a position u_i determined according to the user's commands (u_i is placed at distance d_{u_i} from robot's position and rotated by 45° in the corresponding direction). σ_u represents the concentration factor and was empirically set (see Appendix). If the user does not deliver any command, the contribution of the *policy* is null—namely, all the locations are equiprobable.

It is worth noticing that the implementation of the *user's input policy* particularly matches the discrete control modality that has been exploited in this work. However, other modalities

may require the redefinition of this *policy* (e.g., in the case of continuous control [34]).

D. Direction Policy

The motion of a mobile robot appears more natural when sudden changes of direction are avoided. We defined the *direction policy* to favor the current direction of the robot during the navigation as reported in Fig. 2.

Let ψ be the angle generated by connecting each position around the robot with the current robot position, θ the robot's orientation, the *direction policy* p_d is defined as

$$p_d = \frac{1}{\sqrt{2\pi\sigma_d^2}} \cdot e^{-\frac{(\psi - \theta)^2}{2\sigma_d^2}} \quad (4)$$

where $\sigma_d > 0$ is the dispersion factor controlling the spatial influence of the *policy* (see Appendix VI).

E. Fusion of Policies

We identified the location of the *subgoal* by fusing the output of the aforementioned *policies*. The fusion is responsible for discriminating the most probable zones considering the joint probability of multiple simultaneous events (one per each *policy*). In detail, given a set of N *policies*, we computed the product of the resulting probability grids p_k with $1 \leq k \leq N$

$$\Sigma_p = \prod_{k=1}^N p_k. \quad (5)$$

The result is then normalised to obtain again a probability grid $\hat{\Sigma}_p$, calculated on each position (x, y) in the area around the robot. The location of the *subgoal* (x_{s_g}, y_{s_g}) is identified by

computing the maximum probability of the resulting grid

$$(x_{s_g}, y_{s_g}) = \operatorname{argmax}_{x,y} \hat{\Sigma}_p. \quad (6)$$

In the case of multiple maximum values, the farthest position from the robot is selected. We set the next robot orientation θ_{s_g} along the line connecting the current position and the selected one (x_{s_g}, y_{s_g}) . Finally, the pose $(x_{s_g}, y_{s_g}, \theta_{s_g})$ is provided as input to the motion planner in order to compute the best trajectory the robot will follow to move toward the *subgoal*. Some representative examples are shown in Fig. 2 where typical robotic situations are provided (free space, door passage, corridor, crossroad, and area covered by obstacles). We displayed the results of each individual *policy* (middle panels in the figure), the output of their fusion and the selected *subgoal* (right panels).

F. Refresh of the Subgoal

The frequency of updating the *subgoal* is a crucial aspect. On the one hand, the selection of the *subgoal* has to be as coherent as possible to the environment, to possible unexpected events (e.g., moving obstacles), and to the user's commands. On the other, the higher is the frequency of refresh and the higher is the computational burden. By considering these aspects, a new *subgoal* is computed each time one of the following events occurs.

- 1) The user delivers a new command.
- 2) The robot is close to the *subgoal*. This case represents the condition to force the refresh of the *subgoal* to take into account the new knowledge acquired by the robot.
- 3) A sudden event happens that makes the robot fail in moving toward the *subgoal*; for instance, these situations can be caused by a dynamic obstacle the robot has not previously sensed.

III. MATERIAL AND METHODS

A. Participants

Thirteen healthy users took part in the experiment (S1-S13, 28.9 ± 5.2 years old, 5 females). Users did not have history of neurological or psychiatric disorders and they were not under any psychiatric medication. Nine subjects did not have any previous experience with MI based BMI and four already participated in other BMI experiments (S1, S3, S7, and S9). All participants did not have any previous experience with BMI driven mobile robots. All participants signed a written informed consent in accordance with the principles of the Declaration of Helsinki. The study was approved by the Cantonal Committee of Vaud (Switzerland) for ethics in human research under the protocol number PB_2017-00295.

B. Brain-Machine Interface System

We exploited a classical BMI based on 2-class motor imagery (i.e., kinesthetic imagination of both hands versus both feet) to control the turning direction of the mobile robot. Electroencephalography (EEG) signals were recorded with an active 16-channel amplifier at 512 Hz sampling rate (g.USBamp, Guger Technologies, Graz, Austria), filtered within 0.1 and 100 Hz,

and notch-filtered at 50 Hz. Electrodes were placed over the sensorimotor cortex (Fz, FC3, FC1, FCz, FC2, FC4, C3, C1, Cz, C2, C4, CP3, CP1, CPz, CP2, CP4) according to the international 10–20 system layout. Electroencephalography (EEG) signals were spatially filtered by Laplacian derivation [35] and power spectral density (PSD) was computed via Welch's periodogram algorithm from 4 to 48 Hz (with 2 Hz resolution) in 1 s sliding window every 62.5 ms. The most discriminative channel-frequency pairs (namely, features) were selected during the calibration phase (Section III-C) and used as input to train a Gaussian classifier with a gradient-descent supervised learning approach [4], [18], [36]. In the evaluation phase, features were classified into a probability distribution over the two MI classes with a rate of 16 Hz. Probabilities below 0.55 were rejected. Then, the raw posterior probabilities were integrated over time by an exponential smoothing algorithm [37] and, once a predefined, subject-specific threshold was reached, a discrete turning command (left or right) was delivered to the *shared intelligence* system (Section II).

C. BMI Calibration and Evaluation

The experiment was organised in two recording sessions. On average, the sessions lasted 90 and 112 minutes, respectively (from 1 to 13 days in between). During the first session, participants performed three *calibration* runs (60 trials, 30 per class). The data were used to train the subject-specific BMI decoder. Afterward, each participant completed between three and five *evaluation* runs (40–60 trials, 20–40 per class) over the two recording days to test the BMI performances. In both phases, participants were sitting in front of a monitor and they were instructed by symbolic cues about the motor imagery task to be performed. During *calibration*, they always received a positive visual feedback while in the *evaluation*, they were actively controlling the movement of the feedback through the BMI (Fig. 3(a)). Five subjects (S2, S7, S8, S9, and S11) needed to recalibrate the classifier on the second day.

D. Robot Teleoperations

We asked participants to evaluate our *shared intelligence* system by teleoperating the mobile robot in an office environment (5.40 m $\times$ 8.50 m) with challenging passages, furniture, and open spaces [see Fig. 3(c)]. The navigation task consisted in following a path across seven predefined targets positions (T1-7). A confidence interval of 0.5 m was set to consider a target successfully reached. Two different timeouts were defined for targets T1 and T6 (240 s) and T2-5 (300 s). If a target was missed, the robot was manually moved to the next target position and the user was asked to continue the task from there. Subjects were instructed to perform the navigation task in the shortest possible time. During navigation, participants did not have a direct view of the experimental field and they teleoperated the robot only via the video stream from the on board camera¹.

¹An illustrative video of the experiment is available at <https://youtu.be/ZJ0pckohQ7Q>

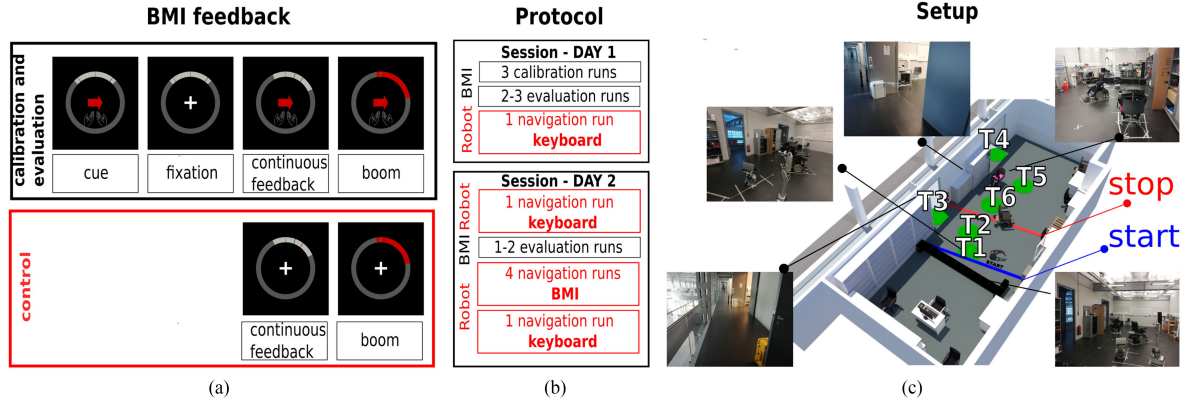


Fig. 3. (a) BMI visual feedback. During BMI calibration and evaluation runs, users were instructed with a visual signal (cue) about the mental task to perform. During the navigation, no cue was provided, and they voluntarily decided the command to deliver. In both cases, they received a continuous feedback representing the output of the BMI decoder. (b) Experimental protocol. In the first session, data to model the BMI decoder was recorded (calibration runs). The BMI was assessed in the following evaluations runs. Finally, participants teleoperated the robot using the keyboard in one navigation run. During the second session users performed two BMI evaluation runs intertwined by three navigation runs using the keyboard and four with BMI. (c) Experimental environment. It included seven target positions (T1–T7). T1 and T2 were placed in free area, T3 and T4 included the passage through two doors, T5 indicated a crossroad, T6 was put in the middle of an area covered by obstacles, and T7 consisted in overstepping the finish line. The user watched the stream from robot's camera.

E. Experimental Conditions

We evaluated our *shared intelligence* system in two teleoperated modalities based on BMI and keyboard input interface. The rationale was to compare the performance of the system with an uncertain control channel (BMI modality) and an optimal and reliable input interface (keyboard modality). In the former, we remapped the two MI tasks decoded by the BMI to the user's intentions of making the robot turn to the left or the right. In the latter, the turning commands were provided by the users by pressing two keys on the keyboard.

Each participant performed four repetitions of the BMI teleoperation condition (during the second session) and three repetitions of the *keyboard* condition. The first keyboard teleoperated run was performed during the first session to make the user get familiar with the environment and the *shared intelligence* system. Overall, subjects completed 91 navigation runs (637 targets in total).

Finally, we compared trajectories in both modalities to an ideal trajectory achieved in a separated experiment where the robot has been teleoperated via joystick (14 repetitions).

F. Mobile Robot and Navigation System

Our robot platform is Robotino by FESTO AG (see Fig. 1). For obstacle detection, we equipped the robot with a laser range finder (Hokuyo URG-04LX-UG01, range 240°, distance 5600 mm) at the height of 53.5 mm and a 3-D camera (Microsoft Kinect v1, field of view 43° vertical x 57° horizontal) mounted at the height of 490 mm. The RGB stream of the camera was shown to the user for telepresence purposes. We exploited an additional laptop (Dell XPS15 9570, Intel Core i7-8750H 2.20 GHz, 16 GB RAM) to overcome the limited computational power of the embedded PC. The laptop was placed on a custom structure fixed on the top of the robot chassis and connected to the robot itself via Ethernet interface.

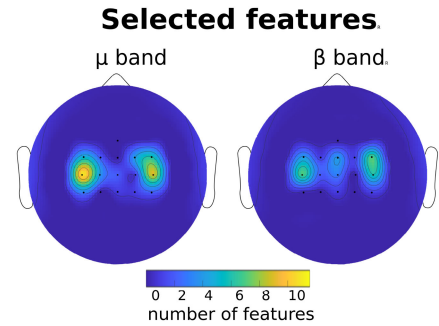


Fig. 4. Topographic representation of the most selected features during the calibration phase for μ and β bands.

We developed our *shared intelligence* system within Robot Operating System (ROS)² ecosystem. We exploited the layers provided by the *grid map package* [38] to store the probability distribution computed by each *policy* and to compute the location of the *subgoal*. Once the new *subgoal* had been defined, we exploited the traditional *navigation_stack* in ROS to plan the trajectory and to move the robot toward the *subgoal*.

IV. RESULTS

A. Initial BMI Decoder Performances

Before evaluating the *shared intelligence* system, we assessed the level of accuracy of the BMI control modality. During the evaluation phase, on average participants achieved an accuracy of $69.30 \pm 10.60\%$, and they were able to deliver a command in 4.91 ± 1.11 s. Furthermore, Fig. 4 shows the spatial and spectral consistency of the features exploited in the BMI decoder across participants. The most frequent activated channels are C3 and C4 in μ band (11 and 9 times respectively versus 2 times for Cz) and in β band (7 and 6 times respectively versus 3 times for Cz). In addition, channels FC4 and FCz appear in β band (7 times

²[Online]. Available: <https://www.ros.org/>

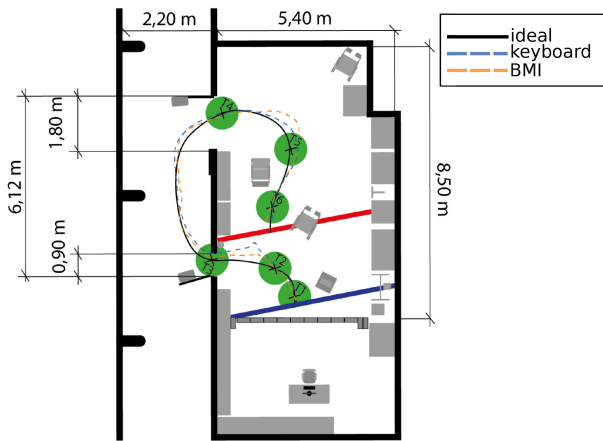


Fig. 5. Average trajectories achieved in the robot navigation for the two control modalities: with keyboard (in blue) and with BMI (in orange). The ideal trajectory is reported in black.

respectively versus 4 for FCz). These results are coherent with the brain cortical regions involved during both hands and both feet motor imaginary tasks [39], [40].

B. Navigation Performance

We evaluated the navigation performance of our *shared intelligence* system according to four objective metrics: 1) percentage of reached targets; 2) distance to the ideal trajectory; 3) time to reach the targets; 4) the number of commands delivered by the user.

First, the proposed *shared intelligence* system allowed to complete the navigation task in both control modalities. Participants achieved 100% of success rate using the keyboard (optimal and reliable interface) and $79.94 \pm 8.32\%$ in the case of BMI. Average trajectories are similar in both conditions (see Fig. 5) as well as the heat maps of the trajectories followed by all participants (Fig. 6(a) and (b)). Differences are mostly related to specific demanding segments (e.g., T3–T4, T4–T5) where an optimal control was required to follow the ideal trajectory.

The similarity between the two control modalities is confirmed by the quantitative metrics reported in Fig. 6(c)–(f). The average distance to the ideal trajectory is 0.24 ± 0.19 m for keyboard and 0.29 ± 0.25 m for BMI control modality. No statistical difference has been found between these two distributions ($p = 0.11$, two-sided Wilcoxon rank-sum tests). Similarly, the length of the path and the time needed to accomplish the task are comparable in the two modalities (2.67 ± 2.71 m versus 2.74 ± 2.95 m, $p = 0.41$; 33.68 ± 33.49 s versus 34.89 ± 38.75 s, $p = 0.29$). Interestingly, only the number of commands required to reach a target is significantly less in the case of BMI modality (4.88 ± 8.59 versus 3.99 ± 6.66 , $p < 0.01$).

C. Level of Assistance

We further analyzed how the reliability of the control interfaces has affected the navigation performances of our system. As expected, a reactive and fully reliable interface such as the keyboard led to a 100% success rate in accomplishing the task

and reaching the different targets. Notably, in the BMI control modality, the navigation results outperformed, on average, the accuracy of the interface computed just before starting the teleoperation task (79.94% versus 69.30%). More in detail, Fig. 7(a) reports the comparison between the reliability of the BMI interface and the quality of the navigation (in terms of distance to the ideal trajectory) suggesting the existence of a slight correlation ($r = -0.21$, $p = 0.49$). However, this can not fully explain the improvement of performances during the BMI navigation task as reported in Fig. 7(b). Indeed, 11 subjects report an increment in the navigation performances, and only the two with already the highest BMI accuracy (S4 and S9) experienced a slight reduction while teleoperating the mobile robot. These results suggest an active role of the *shared intelligence* system in improving the navigation outcomes despite the usage of an imperfect control modality.

Furthermore, the level of assistance provided by the *shared intelligence* system becomes evident if we analyze the way the system implemented (and contextualized) the user's commands. Fig. 8 shows the effective turning angles of the robot after the user's command. On the one hand, in the case of the keyboard (Fig. 8(a)), user's commands are almost directly implemented in the corresponding turning directions (left or right), and the resulting distributions are clearly separated (0.35 ± 0.63 rad and -0.61 ± 0.56 rad for left and right, respectively). On the other hand, during BMI control (Fig. 8(b)), a large amount of user's commands have been discarded by the system (most of the turning angles are close to 0). This fact is due to the ability of the *shared intelligence* system to prevent the implementation of those commands that were accidentally delivered by the BMI, and that would have led to collisions or undesired robot's behaviors. It is worth noticing that the rest of the commands were correctly implemented by the system to let the robot turn left or right. Fig. 8(c) illustrates an example of the contextualization of the user's command in different situations (i.e., corridor, door passage, and collision avoidance for subject S13).

D. Human Evaluation

At the end of the second day, we administrated a questionnaire to assess the subjective evaluation of our *shared intelligence* system. The questionnaire was composed of six questions reported in Table I and related to the two control conditions (keyboard versus BMI). Participants were asked to answer to the questions according to a 5-point Likert scale between two bipolar adjectives. Results are reported in Fig. 9.

Generally, our system was assessed positively in both modalities. As we expected, keyboard was an easier control modality (Q1, 4.15 ± 0.90 versus 2.31 ± 0.95 , $p < 0.001$) that required less workload than BMI for the users (Q2, 1.77 ± 0.83 versus 4.15 ± 1.07 , $p < 0.001$). Nevertheless, in both modalities the robot's behavior was considered quite natural (Q3, 3.23 ± 1.09 versus 2.69 ± 0.87 , $p = 0.119$) and it was appreciated (Q4, 3.54 ± 1.20 versus 2.62 ± 1.03 , $p = 0.061$). Furthermore, Q5 and Q6 confirmed that the behavior was closed to participants' expectations (Q5, 3.46 ± 0.97 versus 3.15 ± 0.99 , $p = 0.46$) and it reflected their expectation, especially in the

Comparison between keyboard and BMI

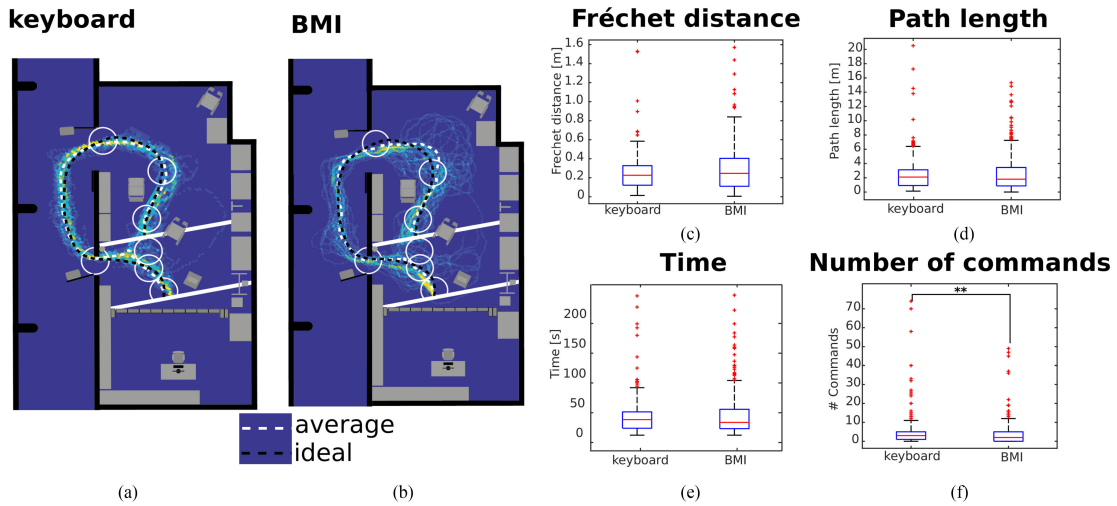


Fig. 6. Comparison of the navigation performance in the two input modalities: keyboard versus BMI. (a) and (b) Heat maps of the trajectories performed by the robot. Maps resolution is 5 cm. The color palette varies from blue (low) to yellow (high hit rate). The start and the finish line (T7) are reported in white, targets T1-6 are represented with white circles. The average trajectory for the keyboard and BMI navigation are shown with white dashed line, while the ideal manual one in the black dashed line. (c)–(f) Distribution of the Fréchet distance to the ideal trajectory, the length of the path, the time to accomplish the task, and the number of delivered commands. Boxes represent the 75th and 25th percentiles and the horizontal line the median. Whiskers extend to the largest and smallest nonoutlier values. Outliers are marked with red crosses. Statistically significant differences are shown with two-sided Wilcoxon rank-sum (**): $p < 0.01$.

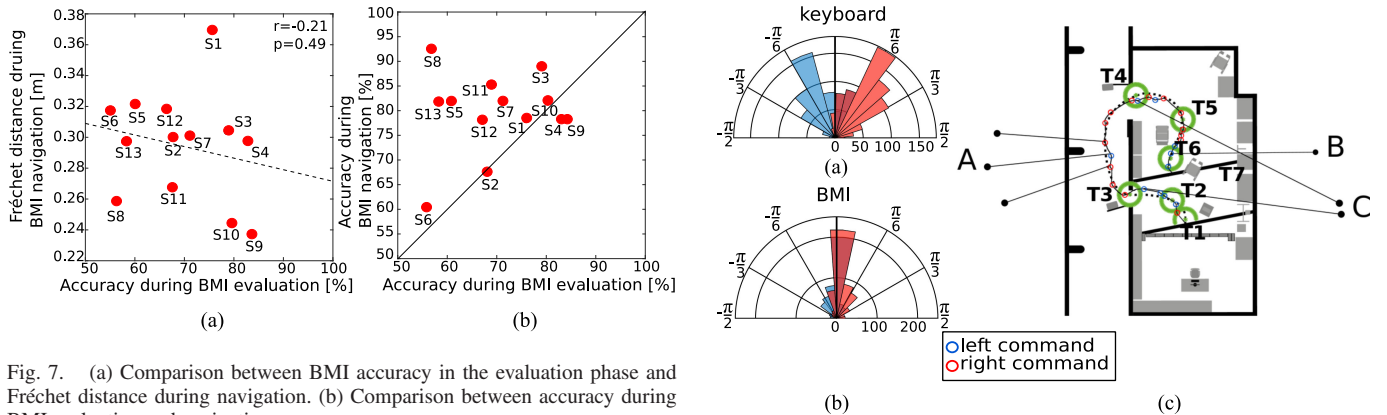


Fig. 7. (a) Comparison between BMI accuracy in the evaluation phase and Fréchet distance during navigation. (b) Comparison between accuracy during BMI evaluation and navigation.

case of the keyboard control modality (Q6, 4.23 ± 0.73 versus 2.54 ± 0.77 , $p < 0.001$).

V. DISCUSSION

This work aims at proposing a new approach for integrating user's intentions and robot's intelligence for BMI teleoperated mobile robots. In literature, research mostly focused on interaction strategies that exclusively favourite the role of the user or the autonomy of the robot [8], [12]–[14], [41]. Although there exist approaches that try to equally fuse the human intentions and robotic decisions [4], [16], [42], they are scarce and often not fully investigated. In this work, we have proposed a novel *shared intelligence* system where human's and robot's intelligence are concurrently contributing to the decision process during BMI teleoperated navigation tasks. The system has been designed so that heterogeneous sources of information (e.g.,

Fig. 8. Distribution of the robot rotations at the occurrence of left and right user's commands in keyboard (a), and BMI. (b) modalities. (c) Illustrative example showing the different robot's behaviors emerged in the *shared intelligence* according to the specific situations. Despite the user's commands, the robot implemented only small changes in direction when navigating in a corridor (A) or passing through a door (B). In (C) the robot discarded the user's command to avoid collisions with an obstacle.

user's commands, position of obstacles, salient locations in the environment) equally define the next robot's action (*subgoal*) through a flexible, easy-to-design and environment-independent set of *policies*.

A. Teleoperations Through Shared Intelligence

The first outcome of this work shows that the proposed approach guarantees an effective interaction strategy between the user and robotic device and allows safe and successful teleoperations in a crowded office environment.

TABLE I
NAVIGATION QUESTIONNAIRE

	Question	Scale (1-5)	
Q1	a. Was the robot easy to control with keyboard?	Very hard	Easy
	b. Was the robot easy to control through BMI?		
Q2	a. How much effort did it require to control the robot with keyboard?	Almost none	A lot
	b. How much effort did it require to control the robot through BMI?		
Q3	a. Do you think the robot's behavior was natural when driven by keyboard?	Not at all	Totally
	b. Do you think the robot's behavior was natural when driven by BMI?		
Q4	a. Did you like the robot's behavior driven by keyboard?	Not at all	Totally
	b. Did you like the robot's behavior driven by BMI?		
Q5	a. Did you expect this robot's behavior driven by keyboard?	Not at all	Totally
	b. Did you expect this robot's behavior driven by BMI?		
Q6	a. How much the robot's behavior driven by keyboard was in line with your intention?	Not at all	Totally
	b. How much the robot's behavior driven by BMI was in line with your intention?		

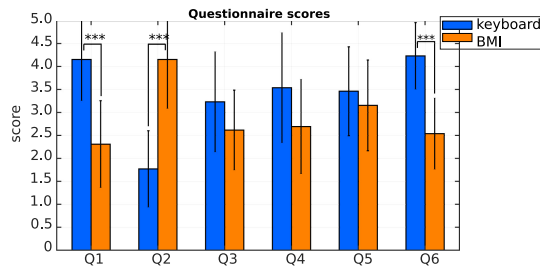


Fig. 9. Results of the qualitative questionnaire for the keyboard (in blue) and for the BMI (in orange) control modality. Mean and standard error are reported. Statistically significant differences are reported with two-sided Wilcoxon rank-sum tests, (***): $p < 0.001$.

Such performances have been achieved with a reliable—although limited to two discrete turning commands—control interface as the keyboard, but also with a demanding and sub-optimal interface such as the BMI. In this control modality, a high mental effort was required to the user and, in addition, the intentions might have been misinterpreted by the BMI decoder, and thus, wrongly delivered to the robot (see Section IV-A). Nevertheless, the proposed *shared intelligence* approach did not only facilitate the teleoperation via BMI, but also it allowed users to achieve a level of navigation performances similar to the most reliable control modality via keyboard.

Finally, results from the subjective evaluation further confirmed that the system was generally well accepted by participants.

B. Role of Shared Intelligence

A full analysis on the influence of the proposed approach would have been required to evaluate all possible experimental conditions, namely the navigation performances in both control modalities (keyboard and BMI) without the active support of the

shared intelligence system and also the validation with respect to other state-of-the-art approaches. However, the burden for the participants would have been excessive, and it would have introduced confounding factors such as fatigue, lack of attention, and loss of interest. Furthermore, a large body of literature has already compared the performances of brain-actuated devices before and after the integration of some level of robotic assistance [8], [17], [25], [43], [44]. On these grounds, in this work, we have directly focused on the evaluation of the *shared intelligence* system with a particular angle on understanding the contribution of the user's and the robot's intelligence in the decision process.

To discuss the *shared intelligence* contribution, first, we have quantified the reliability of the BMI control modality during the evaluation phase (e.g., when the robot is not involved). It is worth mentioning that the BMI evaluation accuracy was the only direct metric to assess the interface reliability in the absence of cues that are necessary to formally derive command accuracy during navigation. On average, results showed sub-optimal reliability of the BMI and scarce reactivity especially if compared to the perfect keyboard modality, but perfectly in line with other previous studies [19], [45], [46]. Nevertheless, the presented *shared intelligence* system was able to correctly interpret (and implement) BMI commands as substantiated by the negative trend in the correlation between BMI accuracy and quality of navigation. At first glance, results suggest that the *shared intelligence* system was robust enough not only to correctly contextualize user's intentions but also to neglect wrong commands accidentally delivered by the BMI.

Evidence of the assistance provided by the *shared intelligence* is further suggested by the consistent improvement in the navigation performances with respect to the quality of BMI control modality. Although we acknowledge that additional factors may influence these results (e.g., controlling a visual feedback on the screen or a real robotic device in a crowded environment is substantially different), we speculate that this is a further indication of the active support provided by the system.

Finally, we have shown that our system supported the user by contextualizing the commands according to the environment. First, a turning command delivered by the user can be implemented in a range of turning directions of the robot according to the different environmental situations.

More interestingly, the system was also able to prevent the actuation of user's commands if they were considered unsafe (due to possible collisions) or leading to undesired behaviors. This was the case in the BMI control modality, where the system assisted the user in successfully accomplishing the navigation tasks by effectively implementing only those commands that were considered correct.

VI. CONCLUSION

In this article, we proposed a *shared intelligence* system that provides a novel interaction strategy for brain-actuated teleoperated mobile robots. The system relied on the idea that user's intent and robot's intelligence are concurrent and equally

TABLE II
SHARED INTELLIGENCE FRAMEWORK PARAMETERS

Configuration Parameters		
Parameter	Meaning	Value
$m \times n$	Size of the neighborhood A	$4 \text{ m} \times 4 \text{ m}$
λ	Weight in the <i>obstacle-avoidance policy</i>	$\frac{1.5}{\log 4 + 1.5 \log 3}$
s_r	Robot size in the <i>obstacle-avoidance policy</i>	0.37 m
$[d_{min}, d_{max}]$	Range in the <i>preferable distance policy</i>	[0.5, 1.5] m
α_{ui}	Target angle in the <i>user input policy</i>	$\frac{\pi}{4}$
d_{ui}	Target distance in the <i>user input policy</i>	1.5 m
σ_{ui}	Target influence in the <i>user input policy</i>	1
σ_d	Target influence in the <i>direction policy</i>	1
d_s	Update distance for the <i>subgoal</i>	$\frac{d_{tot}}{2}$

important factors to define and implement the robot's behavior during navigation tasks.

Thirteen healthy users evaluated the system with two control modalities: via keyboard, a discrete but reliable input interface, and via MI-based BMI. The experiment lasted two recording days and in total consisted of 637 repetitions of the teleoperated navigation tasks. First, results demonstrated that the *shared intelligence* system was successfully supporting the users in accomplishing the task in both control modalities. Moreover, despite the low-bit rate and the uncertainty characterising the BMI, the navigation performances were comparable to the ones achieved with the keyboard. Finally, results highlighted the active role of the *shared intelligence* system in improving the BMI navigation performances and in contextualizing the user's intentions according to different environmental situations.

Future directions will focus on evaluating the influence of the parameters in the system and the consistency of the performance by assigning different weights to the *policies*. From this point of view, we speculate that the best configuration of the *policies* weights might be subject-dependent (e.g., the BMI user prefers keeping more robot's control or needs more assistance) and environment-dependent. In line with this aspect, we plan to add new *policies* in the future to guide the robot behavior and to improve the user's navigation experience. The modular structure of our approach allows to easily design and directly integrate new *policies* in the *shared intelligence* system. For instance, we consider worth introducing a customized set of subject-specific *policies* that can actively support navigation in those situations that are recognized as the most demanding for the user.

Finally, it is worth comparing the performance of our *shared intelligence* approach with other existing shared-control systems in a cross-based experimental setup designed *ad hoc* to test different methods with more groups of BMI users. Moreover, we will extend our *shared intelligence* approach with other assistive platforms such as powered wheelchairs, exoskeletons, and robotic manipulators.

APPENDIX

Table II reports the values of the parameters used to define the *policies* of the *shared intelligence* system. The parameters have been empirically determined by simulation before conducting the final experiments whom results are reported in the manuscript. It is worth mentioning that the parameters have not been tuned nor optimized for the specific environment and

they have been used for all participants, in both experimental conditions and during the whole duration of the experiment.

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